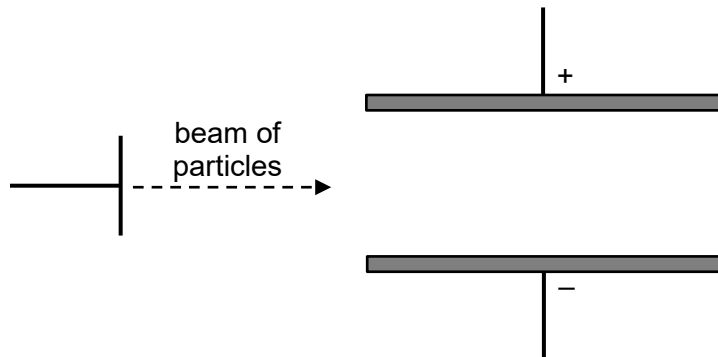


- 2 In two separate experiments, a beam of $^{17}\text{O}^+$ ions and a beam of $[^{16}\text{O}^{18}\text{O}]^{2-}$ ions, travelling at the same velocity, pass through an electric field as shown.



Which statements are correct?

- 1 The $[^{16}\text{O}^{18}\text{O}]^{2-}$ ions are deflected to a larger extent than the $^{17}\text{O}^+$ ions.
- 2 The $[^{16}\text{O}^{18}\text{O}]^{2-}$ ions are deflected in the opposite direction to the $^{17}\text{O}^+$ ions.
- 3 The beam of $^{17}\text{O}^+$ ions travels in a straight path towards the negatively charged plate.

A 1 and 2

B 1 and 3

C 2 and 3

D 2 only